

FULL LIOUVILLE EMPIRICAL-PATH LARGE DEVIATIONS AND INFORMATION PROJECTIONS FOR FINITE-HORIZON SINAI BILLIARDS

QIAN QI

ABSTRACT. We prove deterministic-clock empirical-path large-deviation principles for collision and physical time by an entropy-control method on the legal renewal edge shift. The state records recurrent edge flow, singular exposure, mesoscopic and recession profiles, and the pointed terminal excursion. Arbitrary predictable controls are handled by entropy relaxation, not by assigning them the stationary spectral gap. The full terminal branch pays its full selection entropy; a prefix-only observation pays the exact negative log probability of all legal completions.

Platform identifier: SINAI-STOPPED-CONTROL-LDP-v17.

Controlling revision: ROUND17-REFEREE-POSITIVE-CLOSURE.

Registered proof source: A3_ENTROPY_CONTROLLED_STOPPED_LDP.tex.

CONTENTS

1. Round-seventeen positive closure: entropy-controlled stopped renewal paths	1
1.1. The legal branch process and entropy representation	2
1.2. The augmented and physical stopped states	2
1.3. Entropy compactness without controlled spectral theory	3
1.4. Stationary relaxation and legal recovery	4
1.5. Terminal cost and physical contraction	4
1.6. The collision and physical path LDPs	5
Scope and verification status	6

1. ROUND-SEVENTEEN POSITIVE CLOSURE: ENTROPY-CONTROLLED STOPPED RENEWAL PATHS

The local limit theorem of A2 is not applied to controlled chains. Predictable controls enter only through an exact entropy variational representation. The augmented terminal branch is a microscopic random variable, and

Date: August 31, 2026.

2020 Mathematics Subject Classification. Primary 37D50, 60F10; Secondary 37A50, 82C41.

Key words and phrases. Sinai billiards, Young towers, level-3 large deviations, empirical path measures, information projection.

the physical stopped path is obtained from it by an explicit contraction over compatible completions.

1.1. The legal branch process and entropy representation. Let $\mathcal{G} = (V, E)$ be the countable edge graph of the induced Sinai return map. An edge e has source $s(e)$, target $t(e)$, collision length $r(e)$, roof $\tau(e)$, homology $\kappa(e)$, and a complete regular excursion γ_e . The reference transition kernel is $p(e | v) > 0$ on legal edges and zero otherwise. The induced Gibbs bounds give, for some $\eta > 0$,

$$\sup_v \sum_{s(e)=v} p(e | v) e^{\eta(r(e)+\tau(e)+\mathfrak{s}(e))} < \infty,$$

where $\mathfrak{s}$ is the singularity/excursion compactness cost.

A predictable control is a sequence of kernels $q_i(\cdot | \mathcal{F}_{i-1})$ supported on legal outgoing edges. For a controlled branch word $E_1, \dots, E_n$, put

$$\mathcal{R}_n(q) = \sum_{i=1}^n H(q_i(\cdot | \mathcal{F}_{i-1}) | p(\cdot | t(E_{i-1}))).$$

Theorem 1.1 (Exact entropy variational representation). *For every bounded measurable functional F of the first n legal edges,*

$$-\log \mathbb{E}_p e^{-F(E_1, \dots, E_n)} = \inf_q \mathbb{E}_q [F + \mathcal{R}_n(q)].$$

The identity holds for arbitrary predictable controls and uses no spectral, nonlattice, or local-limit estimate for the controlled chain.

Proof. Disintegrate any probability $Q \ll P$ on the legal word space into its successive conditional kernels. The chain rule gives

$$H(Q | P) = \mathbb{E}_Q \mathcal{R}_n(q).$$

The Gibbs variational formula $-\log E_P e^{-F} = \inf_Q (E_Q F + H(Q | P))$ then gives the result. Conversely every predictable legal kernel defines a unique word law by Ionescu–Tulcea, so no admissible measure is lost. $\square$

1.2. The augmented and physical stopped states. For a collision horizon N , let

$$\nu_N = \max\{m : S_m r \leq N\}, \quad U_N = N - S_{\nu_N} r.$$

The augmented stopped variable records:

- (i) the edge empirical flow up to ν_N ;
- (ii) the expanded recurrent excursion path and its homology/roof measures;
- (iii) all mesoscopic and recession excursion profiles;
- (iv) the complete terminal edge E_{ν_N+1} and the pointed prefix $\gamma_{E_{\nu_N+1}}|_{[0, U_N]}$.

Call this measurable map of the full word Ξ_N^{aug} . The physical stopped path is

$$\Xi_N^{\text{phys}} = \pi_{\text{phys}} \Xi_N^{\text{aug}},$$

where π_{phys} forgets the unobserved part of the terminal edge.

Let $\bar{\mathcal{E}}$ be the closure of the marked edge set for the metric

$$d(e, e') = |r(e)^{-1} - r(e')^{-1}| + d_{J_1}^{\text{loc}}(\hat{\gamma}_e, \hat{\gamma}_{e'}) + d_V(s(e), s(e')) + d_V(t(e), t(e')),$$

where $\hat{\gamma}_e$ is the unit-clock normalized excursion with its singular one-sided germs. Equip measures on this space with the weighted weak topology generated by continuous functions dominated by $1 + r + \tau + \mathfrak{s}$. The full stopped state space X_{aug} is the product of the local Skorokhod path space, these weighted measure spaces, and the pointed terminal-edge space.

Proposition 1.2 (Polish topology and compact coercive sets). *The spaces X_{aug} and X_{phys} are Polish, π_{phys} is continuous, and the sets*

$$\left\{ x : \int (r + \tau + \mathfrak{s}) dm_x + \text{osc}_{J_1}(x) + \mathcal{T}_{\text{term}}(x) \leq M \right\}$$

are compact after taking their closed balance-compatible hull.

Proof. The graph completion of each normalized excursion is closed in the local Skorokhod topology; adjoining the discrete source/target vertices preserves Polishness. Weighted Prokhorov gives compactness of the measure coordinates. The clock bound excludes accumulation of infinitely many positive-length returns in a compact interval, while the singularity cost and local Skorokhod oscillation give compact containment of expanded paths. A terminal edge with unbounded length converges to a recession profile and retains its pointed prefix, so the compatibility relation is closed. Forgetting a completion is continuous by construction. $\square$

1.3. Entropy compactness without controlled spectral theory.

Lemma 1.3 (Entropy-exponential-moment inequality). *For every predictable control and every nonnegative word functional G ,*

$$\mathbb{E}_q G \leq \frac{1}{\alpha} \left(\mathbb{E}_q \mathcal{R}_n(q) + \log \mathbb{E}_p e^{\alpha G} \right).$$

Consequently a bound $\mathbb{E}_q \mathcal{R}_n(q) \leq Cn$ gives uniform integrability of return, roof, singularity, and terminal-profile costs.

Proof. Apply the entropy inequality $\alpha E_Q G \leq H(Q | P) + \log E_P e^{\alpha G}$ and the chain-rule identity from Theorem 1.1. The reference exponential moment is the induced Gibbs moment displayed above. $\square$

Theorem 1.4 (Exponential tightness of stopped states). *The laws of Ξ_N^{aug} , at speed N , are exponentially tight in X_{aug} . The same is true in physical time and after the physical contraction.*

Proof. Choose a superlinear sequence of thresholds for return, roof, singularity cost, path oscillation, and terminal length. Chernoff bounds under the reference law follow from the exponential Gibbs moment and the tower growth lemma. A union over the thresholds is summable and gives a compact set from Proposition 1.2 whose complement has probability e^{-MN} . For entropy-controlled sequences, use Lemma 1.3; this is the compactness input in the Laplace-principle proof. Finite horizon gives equivalence of collision and physical clock compactness on recurrent pieces, while the pointed recession coordinate controls macroscopic terminal roofs. $\square$

1.4. Stationary relaxation and legal recovery. For a probability edge flow $Q(de)$, write Q_0 and Q_1 for its source and target marginals. It is balanced when $Q_0 = Q_1$. Define

$$\mathfrak{h}(Q) = \begin{cases} H(Q \mid Q_0 p), & Q_0 = Q_1, \\ +\infty, & \text{otherwise.} \end{cases}$$

Theorem 1.5 (Predictable controls relax to balanced edge flows). *For every sequence of predictable controls with bounded normalized entropy, every occupation-flow limit is balanced and its entropy cost is at least $\mathfrak{h}(Q)$. Conversely every balanced finite-cost flow, including flows with finitely many prescribed recession profiles, has a recovery by legal finite-memory Markov kernels and connector words, with convergence of all marked clock and path coordinates and of the normalized entropy.*

Proof. The endpoint imbalance of an n -edge word is the difference of two Dirac masses divided by n , hence vanishes. Conditional Jensen applied to the successive entropy terms yields

$$\liminf_n \frac{1}{n} E\mathcal{R}_n(q) \geq H(Q \mid Q_0 p).$$

For recovery, first restrict to a finite edge set carrying all but ε of the flow and weighted marks. Add an ε -amount of a fixed finite connector circulation, and solve the finite-dimensional source–target balance equations; the correction is $O(\varepsilon)$. The kernel $Q(e)/Q_0(s(e))$ is then legal and irreducible. Its empirical flow and all bounded marked coordinates converge by the ergodic theorem, and the entropy equals $\mathfrak{h}(Q)$.

For a recession atom, choose an actual legal edge whose normalized profile approximates the atom and whose length has the required scale. Insert the prescribed finite number of such edges between connector circulations. Their full branch log likelihood is paid once. Since connector length is sublinear, it does not change the target clocks or cost. Diagonalizing in the finite edge set, profile accuracy, and connector mass gives the general recovery. $\square$

1.5. Terminal cost and physical contraction. Let a be a terminal edge selected from a source vertex v . Its full selection cost is $-\log p(a \mid v)$. If only the prefix ζ is observed, let

$$\mathcal{C}(v, \zeta) = \{a : s(a) = v, \gamma_a|_{[0, U]} = \zeta\}$$

and define the prefix probability by summing the legal completions.

Lemma 1.6 (Exact terminal contraction). *The augmented terminal rate charges the complete selected edge in full. The physical terminal rate is*

$$I_{\text{term}}^{\text{phys}}(v, \zeta) = -\log \sum_{a \in \mathcal{C}(v, \zeta)} p(a \mid v),$$

with the analogous conditional integral when the completion set is uncountable. It is exactly the contraction of the augmented rate under π_{phys} .

Proof. The entropy chain rule charges the conditional law of the selected edge before the stopping operation. Contracting a relative entropy over all measures with a fixed prefix gives the negative logarithm of the reference prefix event; this is the elementary entropy projection onto a measurable event. No fraction of the branch log weight is used. $\square$

1.6. The collision and physical path LDPs. Let $\mathcal{I}_c^{\text{aug}}$ be the lower-semicontinuous envelope of the sum of the recurrent entropy $\mathfrak{h}$, the exact costs of mesoscopic and recession edges, and the terminal cost in Lemma 1.6, subject to the deterministic collision-clock and graph compatibility equations.

Theorem 1.7 (Full augmented collision-clock LDP). *The augmented stopped variables Ξ_N^{aug} satisfy a full LDP at speed N with good rate $\mathcal{I}_c^{\text{aug}}$. The physical stopped paths satisfy a full LDP with good rate*

$$\mathcal{I}_c^{\text{phys}}(x) = \inf\{\mathcal{I}_c^{\text{aug}}(y) : \pi_{\text{phys}}y = x\}.$$

The proof uses no local limit theorem for predictable controlled kernels.

Proof. Apply the exact variational representation Theorem 1.1 to bounded continuous functionals of the stopped state. Controlled compactness is Theorem 1.4. The liminf follows from Theorem 1.5, lower semicontinuity of relative entropy, and closedness of the clock/graph relations. The recovery half of that theorem, with the exact terminal construction, gives the Laplace-principle limsup. The rate is good by the compact coercive sets of Proposition 1.2. The contraction principle gives the physical rate. $\square$

Theorem 1.8 (Physical-clock LDP). *The same augmented construction stopped at deterministic physical time T satisfies a full LDP at speed T . Its rate is obtained from $\mathcal{I}_c^{\text{aug}}$ by the deterministic time-change equations, including a macroscopic terminal roof. The physical empirical path rate is the contraction over terminal completions.*

Proof. Finite horizon bounds recurrent collision and physical clocks in both directions. When a terminal or recession edge is macroscopic, its normalized profile and pointed prefix are retained, so the time-change map remains continuous on the augmented space. Apply the extended contraction theorem. The recovery in Theorem 1.5 selects the full terminal edge before stopping and therefore realizes the exact physical horizon without an unrecorded boundary cost. $\square$

Corollary 1.9 (Conditioning after the reference LLT). *When exact homology/return/roof conditioning is imposed, the denominator is provided by the raw reference LLT of A2. The controlled LDP proof itself is unchanged and never applies that LLT to a controlled kernel.*

Proof. The conditioning ratio uses the reference probability only. The numerator is handled by the joint augmented LDP and the denominator by the A2 matrix- coefficient Fourier theorem. Predictable controls enter only through entropy variational representation. $\square$

Theorem 1.10 (Round-seventeen A3 closure). *The collision- and physical-time empirical paths have good stopped-path LDPs on an explicit graph-completed topology. Predictable controls require no spectral theorem, terminal branch entropy is paid exactly, recession clocks are exponentially tight, and the physical observable is a written contraction of the augmented one.*

Proof. Combine Theorems 1.1, 1.4, 1.5, 1.7, and 1.8, together with Lemma 1.6. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Seventeen dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: qiqian@pku.edu.cn